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Measurement of Energy and Energy Meters.

⇒ Energy is the total power ~~develope~~ delivered or Consumed over a time interval, that is,

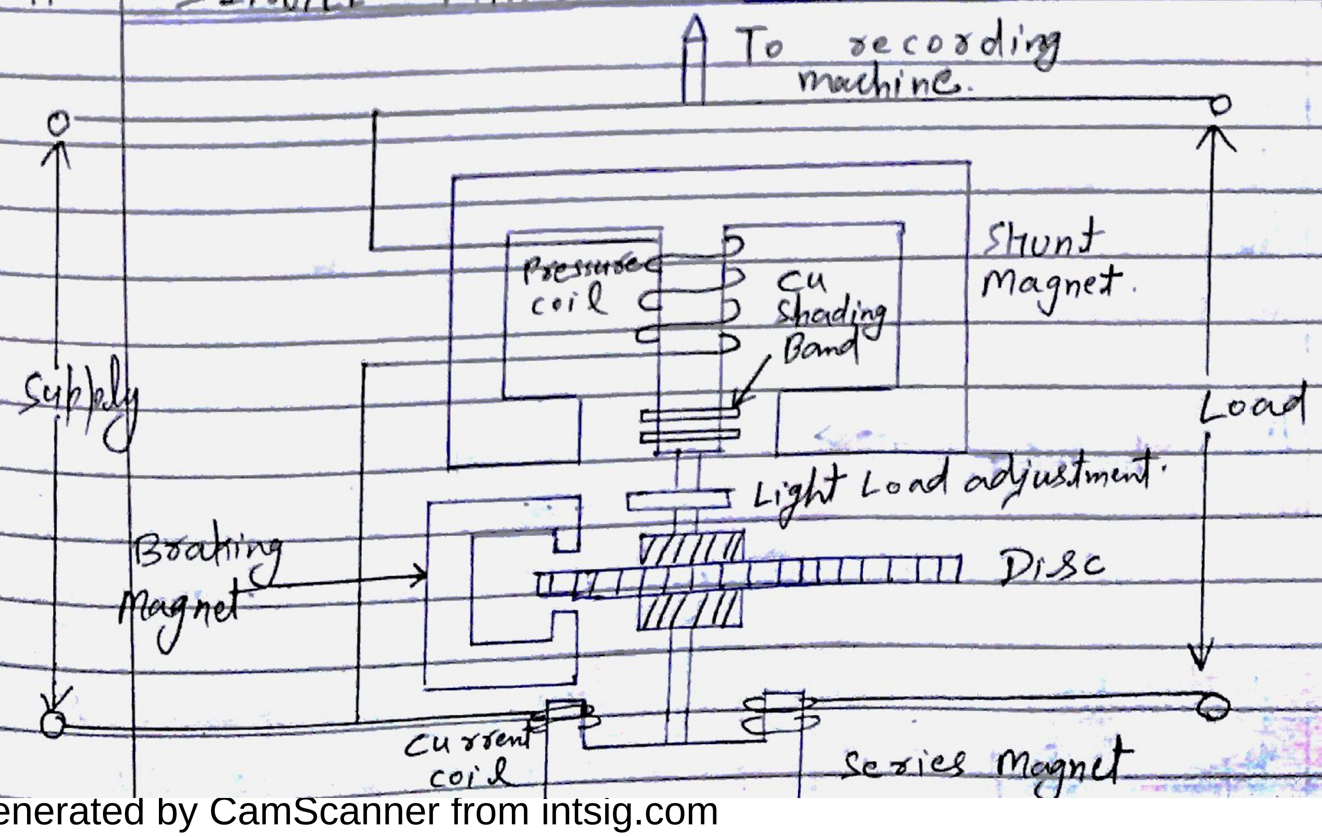
$$\boxed{\text{Energy} = \text{Power} \times \text{time}}$$

Unit → joule or wattsecond or watt hour.

⇒ Electrical energy is measured by means of energy meter (watt-hour meter).

⇒ Energy meter is an integrating instrument.

Single Phase Induction Type Energy Meter



→ Induction type energy meter (single phase) essentially Consist of

(a) driving system (b) Moving system
(c) braking system (d) registering system.

driving System →

→ Consist of two electromagnet
→ Shunt magnet
→ Series magnet.

→ Core is made up of silicon steel lamination.
→ Large number of turns of fine wire is fitted on the middle limb of shunt mag. This Coil is Known as Pressure coil.
→ Each of the two limbs of series electromagnet is wound with few turns of heavy gauge wire. This wound coil is Known as Current coil.
→ The Copper shading bands are also Called the power factor Compensator or Compensating loop.

Moving System →

→ Consist of rotating aluminium disc.

→ Pulsating magnetic field is produced by Shunt electromagnet & it cuts through the rotating disc and indue eddy current there in.

→ Reaction between the two magnetic fields and eddy current set up a driving torque in the disc.

Braking System :→

→ Consist of a permanent magnet, called the brake magnet.

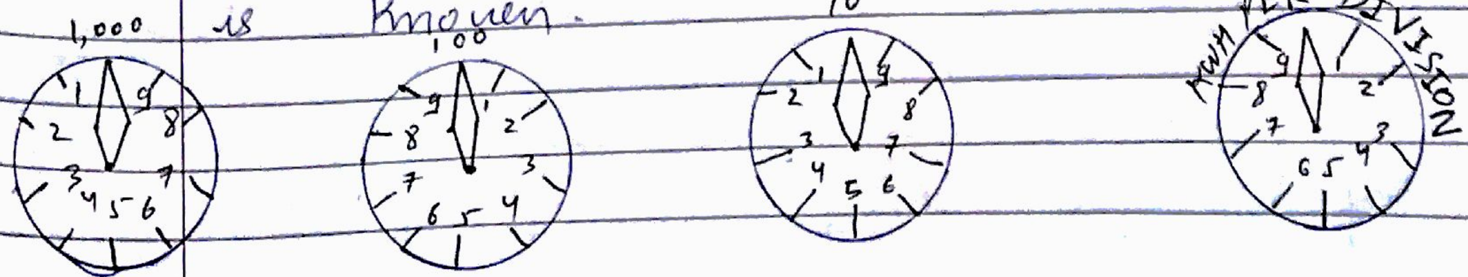
→ disc revolve in the air gap b/w the polar extremities of brake magnet.

→ this rotation set up eddy current in the disc which react with field and exert a braking effect.

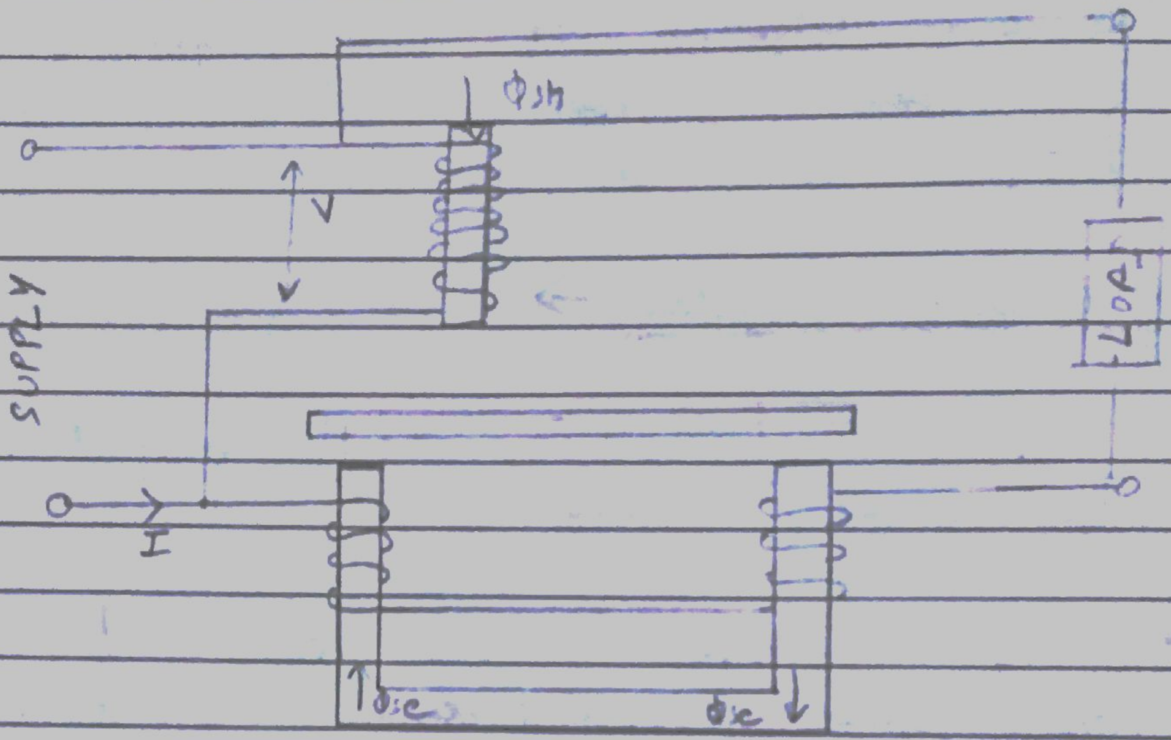
Registering or Counting system :→

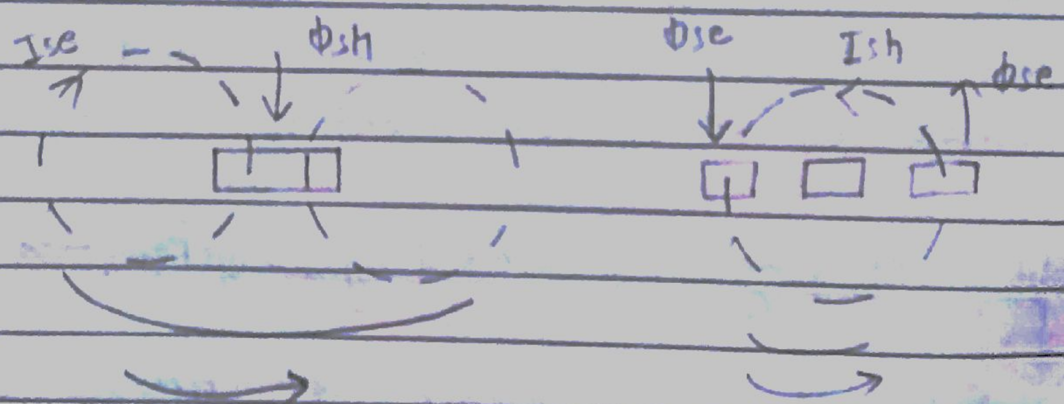
→ Consist of train of gears, driven by either a worm or pinion gear on the disc shaft, which turns pointer that indicate on dials the number of time the disc has turned.

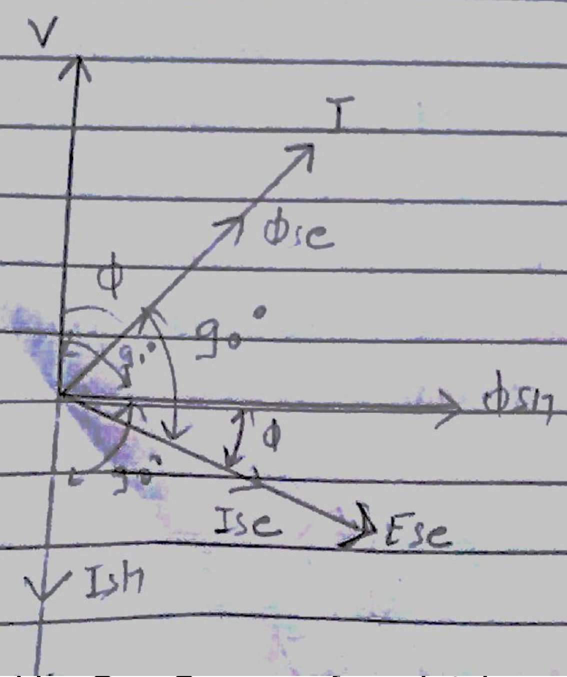
→ Energy meter integrate all the inst. power value so that total energy used over time is knowen.



Theory of operation :-> [Single - φ Induction Type Ene Met]







Phasor diagram

→ Let the load Current lag the circuit voltage V by an angle ϕ.
EMFs E_{sh} and E_{se} are induced in disc by shunt magnet flux ϕ_{sh} & series magnet flux ϕ_{se} respectively & lags behind their respective fluxes by 90°.

→ Now I_{sh} & I_{se} are set up by E_{sh} & E_{se} respectively.
→ Two opposite torque given by $\phi_{sh} I_{se}$ and $\phi_{se} I_{sh}$ will act on the disc. the resultant torque is difference of the two.
→ Phase angle b/w ϕ_{sh} & I_{se} is ϕ
Phase angle b/w ϕ_{se} & I_{sh} is $(180-\phi)$.

Average torque $\propto \phi_{sh} I_{se} \cos\phi - \phi_{se} I_{sh} \cos(180-\phi)$
$\propto (\phi_{sh} I_{se} + \phi_{se} I_{sh}) \cos\phi$

Since $\phi_{sh} \propto V$; $\phi_{se} \propto I$; $I_{se} \propto I$ and $I_{sh} \propto V$.
$\phi_{sh} I_{se} \propto VI = K_1 VI$ and $\phi_{se} I_{sh} \propto VI = K_2 VI$.

So, $T_d \propto (K_1 VI + K_2 VI) \cos\phi$
* $\boxed{T_d \propto VI \cos\phi \propto \text{True power of circuit.}}$

→ Braking torque is due to eddy current & (eddy current) $\propto$ disc speed.
i.e. $T_b \propto N$ ✓.

** <u>Meter Constant</u> :→ The number of revolution made by the

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aluminium disc for 1KWH of energy Consumption is called meter constant

$$\text{Meter Constant } K = \frac{N}{\text{Energy}}$$

$$= \frac{\text{Number of revolution}}{\text{KWH}}$$

Errors, compensations and Adjustments :→

<1> Errors ⇒ In single phase induction type energy meters the errors may be caused by driving system or braking system.

The driving system may cause error due to incorrect magnitude of fluxes, incorrect phase angles and lack of symmetry in magnetic circuit, that makes the meter to creep.

The braking system may cause error due to change in brake magnet strength, change in disc resistance, self braking effect of series magnet flux or abnormal friction of moving parts.

<2.> Compensations and Adjustments :→

1. For low Lagging Power Factor.

2. For Friction or Light Load.
3. For creeping
4. For overloads.
5. Full load.
6. For voltage
7. For Temperature

TRIVECTOR METER →

This is an instrument which measures KVAh also KVA of maximum demand. It consists of a Kwh meter and a reactive kilovolt ampere-hour meter in the common case with a special summator mounted between them. The summator is driven by both the meters through a complicated gearing which makes the summator to register KVAh correctly at all power factor.

→ In Trivector meter five gear system are used.

PHANTOM OR FICTITIOUS LOAD →

→ For testing of meter of higher capacities, phantom or fictitious loading arrangment is employed.

→ The pressure circuit is energised from a circuit of required voltage and current circuit of meter is connected across low voltage supply.

MAXIMUM DEMAND METERS :→

→ The additional Capacity installed in order to take care of this maximum demand involves an investment which would give return only for a portion of time.

→ This is achieved by charging at two tariff at the rate of Rs. x per annum per KW plus Rs y per unit of total consumption.

→ Hence a meter, which records the maximum demand of a Consumer during a particular period. Such a meter is known as maximum demand indicator.

⇒ There are two types of maximum demand meter

→ Wright Maximum Demand Indicator
→ Merz Price Maximum Demand Indicator.

Three Phase Induction Type Energy METER.